LESSON PLAN · SCRATCH (GR 6–8) ADVANCED: ALGORITHMS LAB

Lesson 3: Project: Search Showdown

Module 1 | 50 minutes | Project: comparing algorithms | Grades 6–8 · ages 12–14

Objective: Build an experiment that runs linear and binary search on the same data, counts their steps, and reports which is more efficient.

Success criteria

- I compared two algorithms fairly.
- I organised them into procedures.
- I drew a conclusion from evidence.

Key vocabulary

- **fair test**: Changing only the thing you're comparing.
- **efficiency**: How much work an algorithm needs.
- **benchmark**: A test to compare performance.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)
- Printed project brief

Timed lesson plan (50 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Race (5 min): two volunteers search sorted cards with different methods.
5 min	**Watch & plan**	Play the project video; learners plan on the project brief.
25 min	**Build**	Build-along steps, then Which search fits?. Check in halfway: everyone has something working.
7 min	**Test & peer feedback**	Swap and test each other's work: two stars and a wish.
8 min	**Improve & submit**	Improve one thing; Project: Search Showdown.

Unplugged warm-up

Race (5 min): two volunteers search sorted cards with different methods.

Teaching the idea

Empirical algorithm comparison: same data, same targets, consistent step counting. Learners modularise searches into parameterised procedures and draw evidence-based conclusions.

Analogy: A science fair: fair test, measurements, conclusion.

Live demo: Run both searches for 1, 50 and 100; note when linear wins (target 1!).

Questions to ask

Ask	Listen for
Is binary always faster?	Not for the very first item.
How did you keep the test fair?	Same list, target, counting.
What would happen with 1,000 items?	The gap grows.

Differentiation

Support
Provide both searches as procedures; learners add the experiment.

Stretch
Run all targets 1–100 automatically and average the steps.

Common misconceptions

Misconception	How to fix it
One test proves it	Test several targets.
Binary is always better	Only for sorted data and larger lists.

Assessment

Project rubric (also on the project brief):

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Algorithms	Neither works	One works	Both correct, including not found	Both correct and efficient
Measurement	None	Inconsistent counting	Fair, consistent step counts	Automated tests across many targets
Procedures	None	Partly	Each search in a procedure	Clean, reusable design
Results	None	One target	Table of 5 targets	Averages or graph
Conclusion	None	States a winner	Evidence-based conclusion	Explains when each algorithm is best

Exit ticket
One case where linear search beats binary search.

Homework
Explain your results to someone at home using the “guess my number” game.

Answer key — Kitabash activities

1. Which search fits? (perform · categorize)

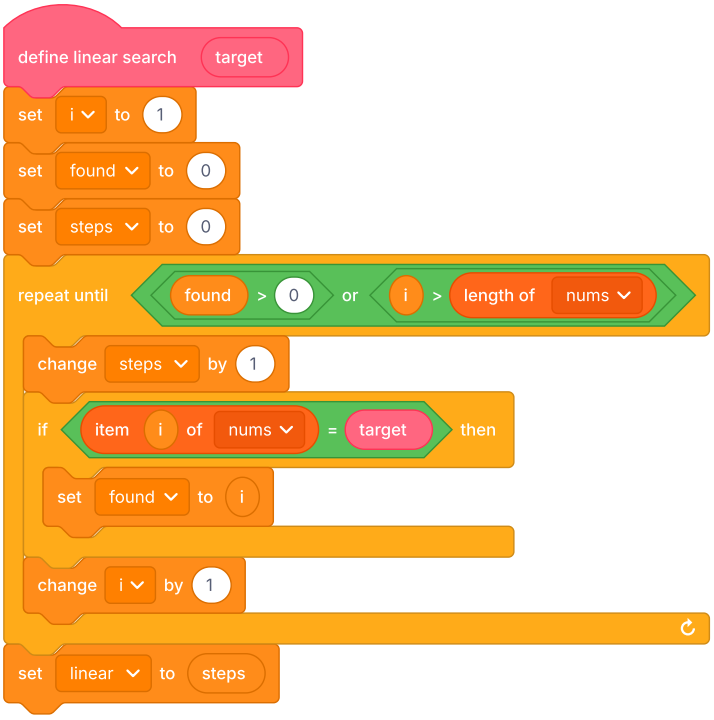
Linear search: A shuffled deck of cards, Today’s 5 unsorted messages, A list you can’t sort
Binary search: A dictionary, A sorted list of 1 million user IDs

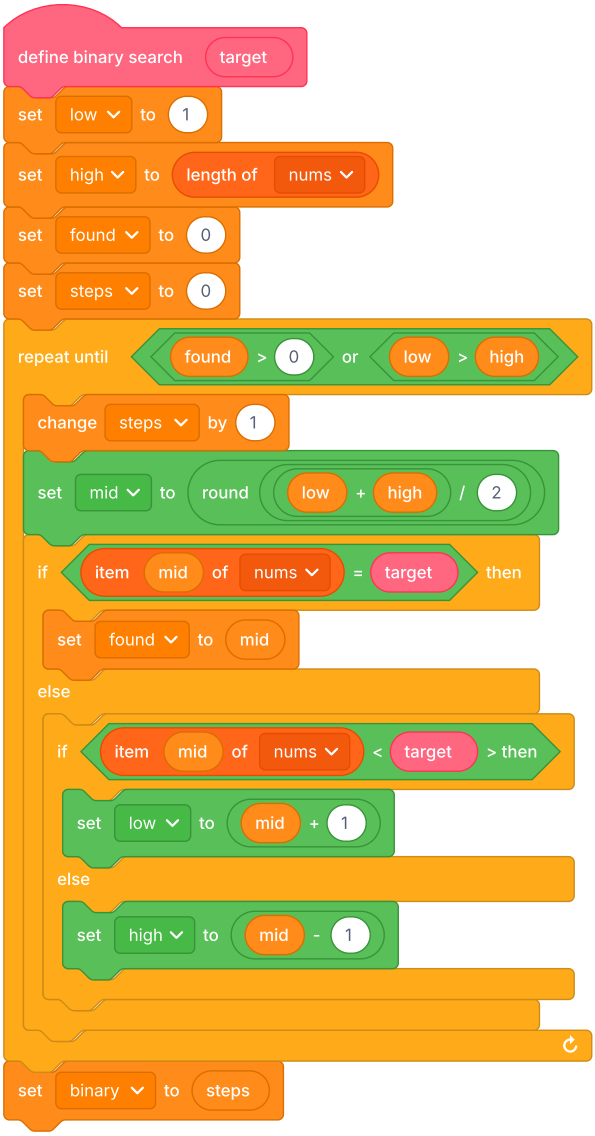
2. Project: Search Showdown (probe · code lab)

Goals checked: Linear count for 90 · Binary count for 90 · Reports both · Uses procedures

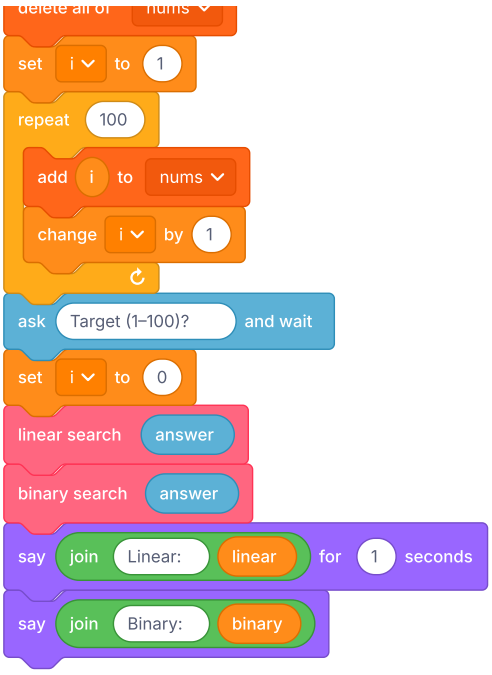
Example project:

Seeker


define linear search target
set i to 1
set found to 0
set steps to 0
repeat until found > 0 or i > length of nums
change steps by 1
if item i of nums = target then
set found to i
change i by 1
set linear to steps


define binary search target
set low to 1
set high to length of nums
set found to 0
set steps to 0
repeat until found > 0 or low > high
change steps by 1
set mid to round low + high / 2
if item mid of nums = target then
set found to mid
else
if item mid of nums < target > then
set low to mid + 1
else
set high to mid - 1
set binary to steps


when clicked



Answer key — worksheet

1. **Key words:** fair test = Changing only the thing you're comparing. · efficiency = How much work an algorithm needs. · benchmark = A test to compare performance.
2. **Which search fits?:** Linear search: A shuffled deck of cards, Today's 5 unsorted messages, A list you can't sort · Binary search: A dictionary, A sorted list of 1 million user IDs
3. **My plan:** A clear plan that covers every requirement.
4. **Checklist:** All requirements met.
5. **Two stars and a wish:** Kind, specific feedback and an improvement.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.